

Alireza Shirzad

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EDUCATION

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|---|-------------------|
| • University of Pennsylvania | Philadelphia, USA |
| • <i>Ph.D. - Computer and Information Science; Advisor: Dr. Pratyush Mishra</i> | 2024 - Present |
| • Sharif University of Technology | Tehran, Iran |
| • <i>M.Sc. - Secure Communication and Cryptography; GPA: 4.00/4.00</i> | 2021 - 2024 |
| • Sharif University of Technology | Tehran, Iran |
| • <i>B.Sc. - Electrical Engineering (Minor: Computer Science); GPA: 3.53/4.00</i> | 2016 - 2021 |

RESEARCH INTERESTS

Applied cryptography, Verifiable Data structures, Verifiable Databases

WORK EXPERIENCE

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| • Lagrange Labs | Remote |
| • <i>Research Intern - Working with Dr. Charalampos Papamanthou and Dr. Dimitrios Papadopoulos</i> | Summer 2025 |

PUBLICATIONS

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- [1] Sebastian Angel, Sydnie-Shea Cohen, Eleftherios Ioannidis, Pratyush Mishra, Alex Ozdemir, Yuxiang Qiu, and Alireza Shirzad. “Zippel: A Language, Compiler, Runtime, and Analyzer for ZK Proof Systems”. In: *IEEE S&P 2027*. Authors listed in alphabetical order.
 - [2] Hossein Hafezi, Alireza Shirzad, Benedikt Bünz, Kevin Lewi, Dillon George, and Joseph Bonneau. “Aegon: Self-Auditable Key Transparency”. In: *Preprint. Hossein and Alireza contributed equally to this work.*
 - [3] Bharath Namboothiry, Alireza Shirzad, Spencer Solit, Ryan Marcus, and Pratyush Mishra. “TruthTable: A Verifiable Query Engine”. In: *Preprint. Also presented at ZKProof VIII and SeQureDB 2026. Bharath and Alireza contributed equally to this work.*
 - [4] Alireza Shirzad, Sriram Sridhar, Dimitrios Papadopoulos, and Charalampos Papamanthou. “Relaxed Modular PCS from Arbitrary PCS and Applications to SNARKs for Integers”. In: *TCC 2026. Also presented at ZKProof VIII.*
 - [5] Hossein Hafezi, Alireza Shirzad, Benedikt Bünz, and Joseph Bonneau. “IronDict: Transparent Dictionaries from Polynomial Commitments”. In: *USENIX Security 2026. Also presented at RWC 2026. Hossein and Alireza contributed equally to this work.*
 - [6] Benedikt Bünz, Tushar Mopuri, Alireza Shirzad, and Sriram Sridhar. “DewTwo: A Transparent PCS with Quasi-Linear Prover, Logarithmic Verifier and 4.5KB Proofs from Falsifiable Assumptions”. In: *CRYPTO 2025*. Authors listed in alphabetical order.
 - [7] Michel Dellepère, Pratyush Mishra, and Alireza Shirzad. “Garuda and Pari: Smaller and Faster SNARKs via Equiefficient Polynomial Commitments”. In: *USENIX Security 2026. Also presented at zkSummit 2025 and SBC 2025. Authors listed in alphabetical order.*
 - [8] Alireza Shirzad. “Designing a Succinct Argument System Based on GKR Protocol via Polynomial Commitment Schemes”. M.Sc. thesis. Sharif University of Technology.

TALKS

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| • Aegon: Self-Auditable Key Transparency | |
| ◦ Meta, Remote | September 2026 |
| • TruthTable: A Verifiable Query Engine | |
| ◦ Yale Applied Cryptography Seminar, New Haven, USA | September 2026 |
| ◦ NYU Cryptography and Security Seminar, New York, USA | May 2026 |
| • IronDict: Transparent Dictionaries from Polynomial Commitments | |
| ◦ USENIX Security 2026, Baltimore, USA | August 2026 |
| ◦ UPenn Security and Privacy Seminar, Philadelphia, USA | October 2025 |
| ◦ Berkeley Crypto Seminar, Remote | October 2025 |
| • Garuda and Pari: Smaller and Faster SNARKs via Equiefficient Polynomial Commitments | |
| ◦ USENIX Security 2026, Baltimore, USA | August 2026 |
| ◦ Science of Blockchain Conference (SBC), Berkeley, USA | August 2025 |
| ◦ SBC 2025 Online Science Fair, Remote | July 2025 |
| ◦ Berkeley Crypto Seminar, Berkeley, USA | December 2024 |
| ◦ CMU LayerZero Crypto Seminar, Pittsburgh, USA | December 2024 |

- Stanford Security Seminar, Stanford, USA *December 2024*
- Sumcheck Builders Club, Remote *November 2024*
- University of Maryland Crypto Reading Group, College Park, USA *November 2024*
- NYU Cryptography and Security Seminar, New York, USA *October 2024*
- UPenn Security and Privacy Seminar, Philadelphia, USA *October 2024*

TEACHING EXPERIENCE

- **Sharif Information and Computer Technology (Sharif ICT)** *Remote*
 - **Blockchain and Web3** *January 2021 - September 2023*

TEACHING ASSISTANT EXPERIENCE

- **University of Pennsylvania** *Philadelphia, USA*
 - CIS 5560: Cryptography *Spring 2026*
 - CIS 5510: Computer and Network Security *Fall 2024*
 - CIS 5560: Cryptography *Spring 2024*
- **Sharif University of Technology** *Tehran, Iran*
 - Linux/Python Lab *Fall 2022*
 - IoT Security *Fall 2022*
 - Object Oriented Programming (Head TA) *Spring 2022*
 - Provable Security *Spring 2022*
 - Numerical Computation *Fall 2021*
 - Object Oriented Programming (Head TA) *Spring 2021*
 - Signals and Systems *Spring 2021*
 - Data Communication Network *Spring 2020*

HONORS AND AWARDS

- Member of the Iranian National Elite Foundation for seven consecutive years - October 2016 to January 2024
- Top 25% of B.Sc. class; granted M.Sc. scholarship, Sharif University of Technology - January 2020
- Ranked 32nd of 200,000 in the Iranian Nationwide Mathematics University Entrance Exam - September 2016

SERVICE

- **Artifact Reviewer**
- 26th Privacy Enhancing Technologies Symposium (PETS 2026)
- **Sub-reviewer**
- ISC International Journal of Information Security
- Financial Cryptography and Data Security 2025
- 46th IEEE Symposium on Security and Privacy (S&P 2025)
- **UPenn Security and Privacy Seminar Coordinator** *September 2024 - September 2025*
- **Schlein Cybersecurity Center Local Seminar Coordinator** *January 2026 - Present*
- **Penn Iranian Academic Network (PIRAN) President** *March 2026 - Present*